Upper Limb Muscle Fatigue Analysis Using Multi-channel Surface EMG

Ahmed Ebied, Ahmed M. Awadallah, Mohamed A. Abbass, and Yasser El-Sharkawy

***Abstract*—Muscle fatigue is a biochemical process that causes several effects, one of them is changes in muscular electrical activity. Electromyography (EMG) signals detect these changes in the form of frequency shift and amplitude variation. In this study, forearm muscle fatigue has been investigated using 8-channel EMG signal recorded from 15 healthy subjects during isometric contraction. We utilise Median Frequency (MDF) and Root-Mean-Square (RMS) to quantify the fatigue effects on frequency and amplitude, respectively. The changes of both (Δ_{MDF}) and (Δ_{RMS}) are the parameters to assess fatigue across subjects and channels. Statistical analysis has been carried out on the results to evaluate the vulnerability of subjects and channels to fatigue. Our methods were able to identify and assess the most and least susceptible subjects and channels to fatigue.**

***Index Terms*—Muscle Fatigue, Electromyography, Median Frequency**

I. Introduction

Muscular fatigue is a decline in performance and losing the ability to maintain or generate force. It is a task-induced phenomenon caused by several biochemical factors accumulated after sustained sub-maximal or maximal contractions. Muscular fatigue can affect the nature and number of activated motor unit (MU) within muscle fibres [1].

These physiological changes can affect the electrical activity of the muscle. Since surface Electromyography (sEMG) is a weighted summation of motor unit action potential propagating along muscle fibres [2]. It is possible to identify and detect muscular fatigue by analysing sEMG [3] which is non-invasive alternative to other fatigue assessment methods such as lactate concentration in blood [4].

Muscular fatigue has been observed to directly affect sEMG by detecting increases in signal amplitude to maintain the required level of force [5]. In addition, fatigue was strongly linked to changes in power spectral density (PSD) of sEMG [6]. It has been established that during maximal contractions PSD shifts towards the lower frequencies. This frequency shift is generally attributed to the decrease in muscle fibre conduction velocity [7], in addition to changes in the number and synchronisation of the recruited MU and the activation of new ones [8].

Therefore, several methods and techniques have been developed in the last decades for muscle fatigue assessment and evaluation [3]. The approaches for fatigue evaluation include time-domain analysis [9], spectral analysis [10], time-frequency analysis (such as wavelet [11]), non-linear approaches such as Entropy and fractal analysis [8]. In addition to Joint Analysis of Spectral and Amplitude (JASA) methods [12]. All of these studies rely on bipolar electrodes for sEMG acquisition.

Recently, multi-channel setup has been employed for fatigue assessment approaches. Multi-channel sEMG provides access to a richer set of relevant variables on the muscle or MU level. This offers several benefits, include the assessment of regional myoelectric manifestations of fatigue [13].

Here, we utilised 8-channel sEMG acquisition system to evaluate forearm muscles fatigue for 15 subjects during wrist's isometric contraction. These muscles have been chosen due to their significance for upper-limb prosthesis control systems which can be deeply affected by fatigue [14]. Amplitude analysis was performed using RMS as a fatigue parameter while MDF was used for the spectral analysis. We aim to introduce a method for identifying the most and least susceptible forearm muscles to fatigue. In addition, to classify the subjects according to their endurance and vulnerability to muscular fatigue.

II. Experimental procedures

A. Participants

Fifteen healthy young male adults (age: 21.56 1.84 years), (mass: 73.55 3.54 kg, height: 177.92 2.58 cm). The inclusion criteria for participation included being physically fit and athletic with no history of neurological dysfunction, and having no injury in the upper limb in the last six months. Prior to the experiment, all subjects were asked to perform a physical test to assess their fitness. All subjects passed the test averaging (74.8 14.13) push-ups and (20.15 5.76) pull-ups.

B. sEMG acquisition setup

The participants were informed about the testing procedure. Then, their skin was prepared by wiping the muscle belly with an alcohol swab. The muscle activity was recorded with Myo-armband (http://www.bynorth.com/). The Myo-armband has 8 medical grade stainless steel sEMG single differential electrodes. Each subject wore the Myo-armband on the forearm. Electrodes were aligned along the palm side of the wrist with *Ch4* placed on *Pronator Teres* muscle as shown in Figure 1. The electrode placement is unified for all subjects to ensure that the numbering of channels the same for all subjects.

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Fig. 1. electrode positioning on the forearm and channel numbers.

Fig. 2. The fatigue protocol. Subjects were asked to hold 6Kg load in 90 angle for 120 seconds.

The Myo-armband sampled 8 sEMG sensors at a 200 Hz frequency with a resolution of 8 bit signed and streamed the data through a *bluetooth* low energy connection to the computer where data was recorded. The sEMG data from the Myo-armband is already notch filtered at 50 Hz to avoid power grid interference. This setup was chosen because it is an affordable option with comparable performance to much more expensive setups such as Delsys Trigno and the Cometa Wave [15].

C. Fatigue protocol

All participants were requested to stand with their elbow flexed to a 90 angle handling a 6 kilograms load using their right hand. Their forearms were positioned horizontally while their hand palm facing up as represented in Figure 2. sEMG were recorded continuously for 120 seconds to reach fatigue.

III. Data Analysis

In this study, two approaches for muscle fatigue evaluation were investigated. Firstly, the Amplitude analysis where RMS was used as a parameter for the temporal and amplitude changes in sEMG because of fatigue. Secondly, the spectral analysis where the fatigue parameter was MDF to detect the frequencies shift and spectral changes. All analysis has been performed using Matlab *(Mathworks, Inc., MA)*.

A. Amplitude analysis

The change of amplitude due to muscular fatigue represents the dominant change of sEMG signal in the time domain [16]. Hence, RMS was used as an indicator for this amplitude modulation, and the change of RMS values with time Δ_{RMS} was a parameter for fatigue assessment.

Firstly, The sEMG envelope for each channel was computed. Then, the signal was segmented into 1 second epochs. Secondly, RMS values were estimated for each epoch following this equation:

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{1}$$

where x_i is the i^{th} sample of a signal and N is the number of samples in the epoch.

To calculate the Δ_{RMS}, the 120 RMS values for the 120 segments were fitted to the first polynomial equation. The slope of this equation is Δ_{RMS}, which would be an indicator of the effect of fatigue on the amplitude modulation. This was performed for each sEMG channel across the 15 subjects.

B. Spectral analysis

MDF is a prominent spectral descriptor that can quantify the PSD. Therefore, the change on MDF with time Δ_{MDF} has been chosen as a spectral parameter for muscle fatigue assessment.

Similar to the temporal analysis, the sEMG of each channel was segmented into 1-seconds epochs. The spectrum for each epoch was computed using Short-time Fourier transform (STFT) to conform to the stationary requirement of the Fourier transform. In addition, a non-parametric approach was used since there are enough data and there was no underlying known structure within the signal.

MDF is defined as the frequency that divides the spectrum in two equal halves, where 50% of the total power within the epoch is reached. For each epoch, MDF was calculated according to the following equation:

$$\int_{0}^{MDF} P(f)df = \int_{MDF}^{f_s/2} P(f)df \tag{2}$$

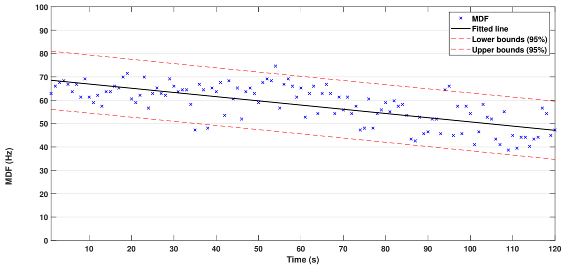

Fig. 3. MDF values for each epoch recorded from Subject 1/ Channel 4. MDF values were fitted to first polynomial equation to calculate Δ_{MDF} for this subject/channel.

where $P(f)$ is the PSD of the epoch and f_s is the sampling frequency.

The MDF value for each epoch was fitted to a first polynomial equation. The slope of this equation is Δ_{MDF}, which was used as a fatigue parameter. An example of the fitted line is shown in Figure 3.

C. Statistical analysis

Both Δ_{RMS} and Δ_{MDF} were computed for each subject and channel to evaluate the muscle fatigue. The slopes across channels have been analysed to determine the most affected channels by fatigue. The subjects slopes have been investigated as well to assess the effect of fatigue on each individual separately.

The slopes (Δ_{RMS} and Δ_{MDF}) were averaged across channels and subjects to evaluate the effects of fatigue on them. Moreover, to test if these changes in slopes are statistically significant, one-sample t-test has been performed to test the null hypothesis that the slopes data come from a normal distribution with mean equal to zero and unknown variance. Hence, by rejecting the null hypothesis, the changes in Δ_{RMS} and/or Δ_{MDF} would be statistically significant.

IV. Results

A. Amplitude results

The Δ_{RMS} for all channels and subjects are represented in a $3D$-bar graph as shown in Figure 4. The Figure shows that Δ_{RMS} values are mostly positive with some outliers. This indicates that sEMG amplitude usually increases gradually with time as fatigue level increases.

B. Spectral results

Unlike Δ_{RMS}, most of Δ_{MDF} values are negative as shown in Figure 5. This indicates that the PSD shifts towards lower frequencies as the fatigue builds up.

C. Statistical results

To investigate the effect of fatigue, the averages of Δ_{RMS} and Δ_{MDF} across subjects and channels were calculated. Figure 6 shows the averages of both slopes across subjects (Panel A and B) and channels (Panel C and D).

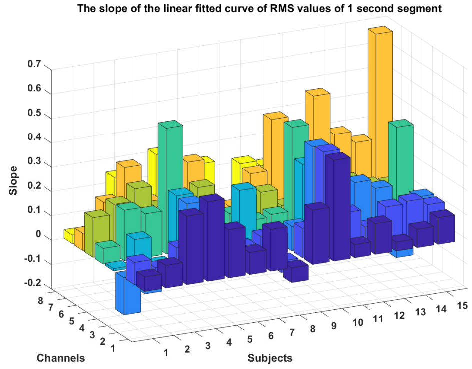

Fig. 4. A $3D$-bar graph for Δ_{RMS} for 8 channels and 15 subjects.

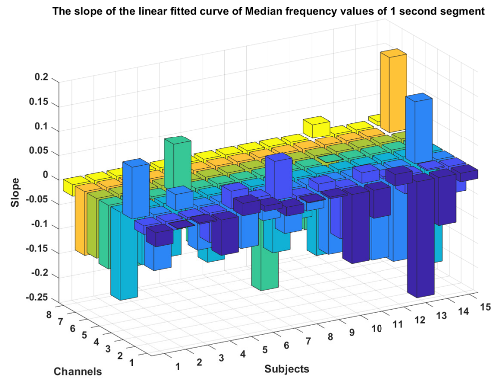

Fig. 5. A $3D$-bar graph for Δ_{MDF} for 8 channels and 15 subjects.

Moreover, the statistical significance of the slopes was explored to assess the effect of fatigue of channels and subjects. One-sample t-test was performed to test the null hypothesis that slopes data comes from a normal distribution with mean equal to zero and unknown variance. The p-values of this t-test are illustrated in Figure 7.

V. Discussion

Multi-channel sEMG signals were used to evaluate and analyse the fatigue process for forearm muscles during wrist's exercise. RMS and MDF were computed to perform amplitude and spectral analysis, respectively, the slopes of both metrics (Δ_{RMS} and Δ_{MDF}) were indicators to the fatigue progression.

The results showed that Δ_{MDF} were mostly negative across channels and subjects as shown in Figure 5. This indicates the PSD shifts towards lower frequencies as less MU are activated as fatigue progress. In this case, the amplitude of recruited MU usually increases to compensate for the inactive MU. Therefore, the RMS of sEMG is expected to increase to keep the same level of force. This was illustrated in Figure 4 with mostly positive Δ_{RMS} across channels and subjects.

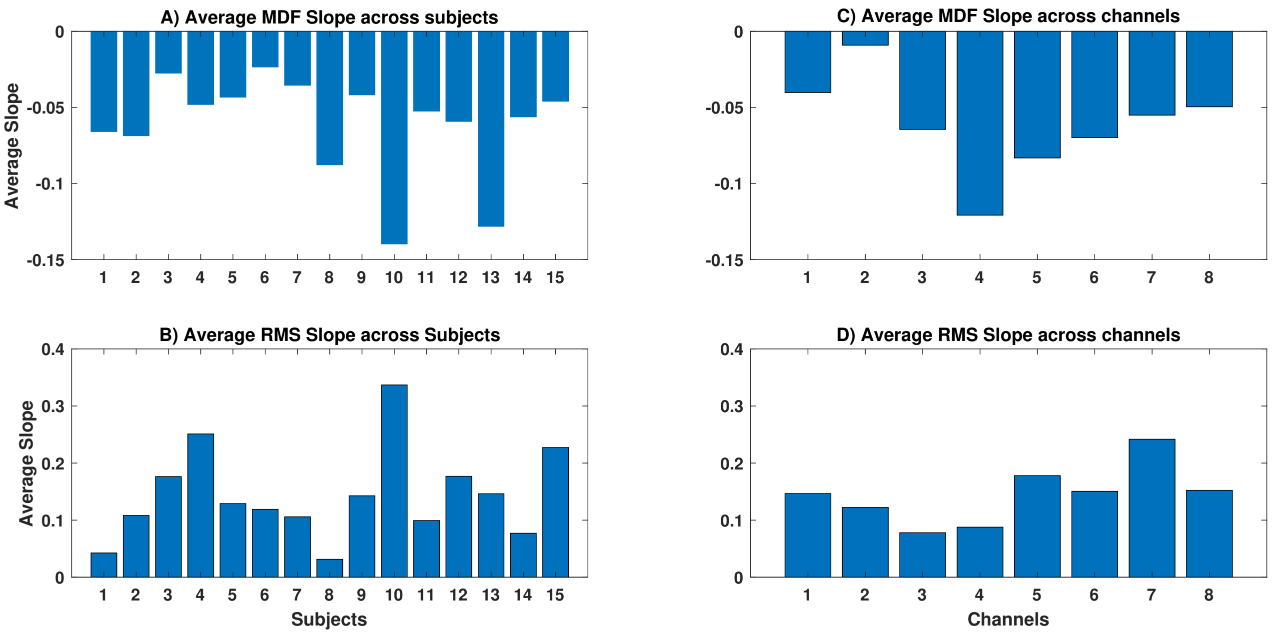

Fig. 6. The average Δ_{RMS} and Δ_{MDF} across subjects and channels. Panels (A) and (C) shows the Δ_{MDF} across subjects and channels respectively. While Panels (B) and (D) shows Δ_{RMS} across subjects and channels as well.

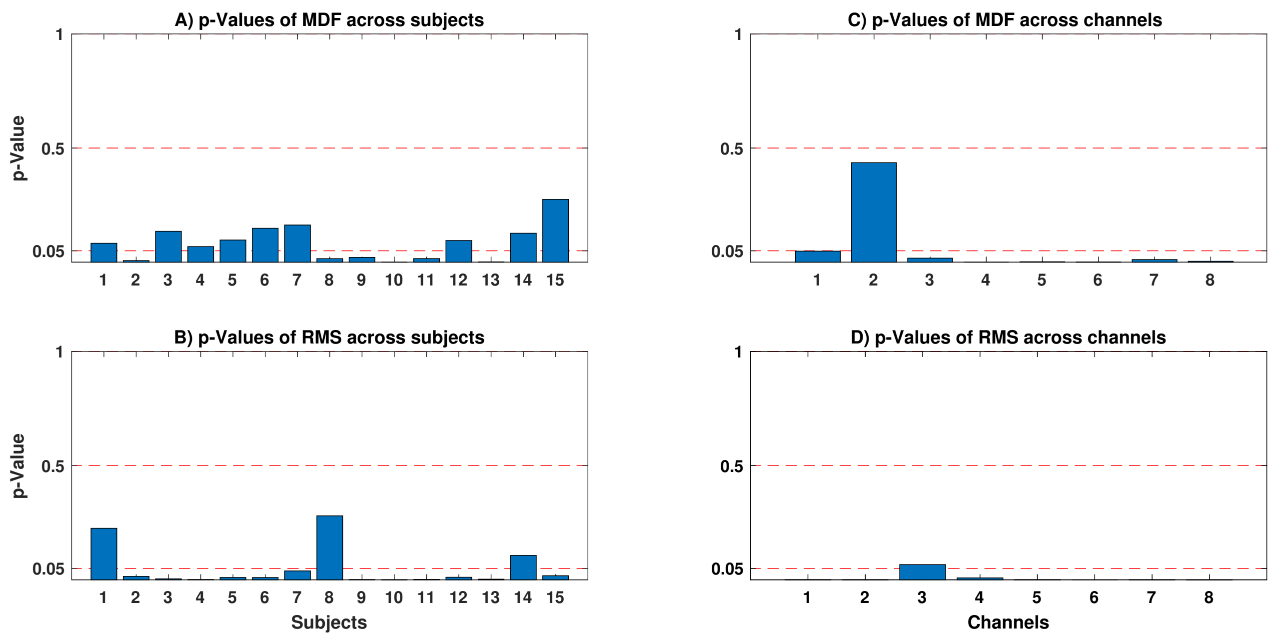

Fig. 7. The p-values of the one sample t-test on Δ_{RMS} and Δ_{MDF} across subjects and channels. Panels (A) and (C) shows p-values across subjects and channels respectively. While Panels (B) and (D) shows p-values across subjects and channels as well.

Statistical fatigue analysis across subjects and channels was interesting. For instance, although average Δ_{MDF} was negative across all channels and subjects, some channels (such as *Ch4*) were more affected than the other as shown in Figure 6.C. On the other hand, *Ch2* has the highest average Δ_{MDF}, hence the frequency shift was minimal in comparison. These conclusions were supported by the one-sample *t*-test as shown in Figure 7.C, where *Ch2* did not reject the null hypothesis ($p \leq 0.05$) demonstrating that the Δ_{MDF} in *Ch2* was not statistically significant. However, changes in RMS across channels were different, *Ch3* was the only channel not to reject the null hypothesis ($p \leq 0.05$) as shown in Figure 7.D. This conforms with the average Δ_{RMS} in Figure 6.D, where *Ch3* has the least average Δ_{RMS}.

The same pattern was found in all subjects. For example, in Figure 6.A, Subjects *Sub3* and *Sub6* had the least frequency shift in comparison with other subjects. In addition, Δ_{MDF} for both subjects did not reject the null hypothesis ($p \leq 0.05$) as shown in Figure 7.A. These results could imply that *Sub3* and *Sub6* were less susceptible to fatigue and had better endurance in general. However, by taking the Δ_{RMS} results into account, we could find that most subjects had a significant change in RMS values except for Subjects (*Sub1*, *Sub8*, and *Sub14*) as shown in Figure 7.B. Therefore, we cannot rely on the MDF results as the final indicator for endurance because of the changes of force levels according to RMS results.

Further work is needed to have a method for identifying subjects with high endurance less vulnerable to fatigue. Although Δ_{MDF} was reliable as a fatigue indicator, Δ_{RMS} should be incorporated in the analysis to control force level and ensure a more precise comparison between subjects. For instance, JASA [12] would be a suitable tool for this target, where both amplitude and spectrum analysed jointly.

VI. Conclusion

To sum up, we investigated muscle fatigue using multi-channel sEMG data using amplitude and spectral analysis across channels and subjects. The results showed the ability to identify channels/muscles affected by fatigue. In addition, further work is needed to improve our approach to identify subjects less vulnerable to fatigue. However, the preliminary results here are encouraging.

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